

This is a strong idea with a clean core claim: **prior work asks whether models abstain enough; you ask whether abstention tracks the evidence or the user.** To reach a top-tier venue, the plan below addresses what reviewers will push on: ecological validity, the "rational updating" objection, whether the model can actually see what humans see, a mechanistic test of "won't say," and a human study powerful enough to support a null claim. It's organized as a calibration study, five core experiments, and the analysis and logistics around them.

Stage 0: Stimulus construction and human calibration

Everything else depends on this, so it deserves the most rigor.

Broaden the stimulus set beyond the synthetic scene. The colored-ball-with-code stimulus is excellent for control, but reviewers will call it a toy. Build three tiers, all with an annotated diagnostic region:

- **Synthetic:** your current scenes, extended to more feature types.
- **Semi-natural:** real photos where one region decides the answer, such as a street sign, a jersey number, a product label, or a car's colour.
- **Natural VQA:** items drawn from TextVQA, GQA, or VQAv2, with bounding boxes you annotate for the diagnostic region.

Cross feature types with spatial frequency. Organize items by what the answer depends on: text or codes (high frequency), colour (low frequency), shape, count, spatial relation, and object identity. Your Fig. 1 already shows that colour survives blur while codes don't. Make this a designed factor rather than a nuisance, since it lets you show that calibration must be per category.

Expand degradation into a 2×2 design. H3 is currently confounded: occlusion both removes evidence and makes the absence visible, while blur corrupts evidence and hides it. Separate the two dimensions:

	Absence visible	Absence hidden
Evidence removed	Grey or patterned occluder	Inpainted removal (region filled plausibly)
Evidence corrupted	Blur with a visible "blurred region" frame or mosaic	Gaussian blur, noise, JPEG artifacts, downsampling

This turns H3 from a comparison of two manipulations into a test of your actual theory: that legibility of absence drives calibration. That is a much stronger contribution.

Run human calibration as a proper psychometric study. Use a 4AFC task on Prolific. First, run QUEST staircases in a pilot to locate thresholds per category. Then use the method of constant stimuli to fit logistic or Weibull psychometric functions per item, or per category with item random effects. Choose the parameter values that hit your anchors (95/80/50/≤35%). Aim for roughly 20–30 ratings per item per level.

Run a second human condition that allows "can't tell." This gives you a normative abstention curve, meaning how often humans abstain at each level. Without it, "calibrated abstention" has no reference point. Also collect confidence ratings so you can report human metacognitive sensitivity alongside the models'.

Calibrate the models too. Human thresholds are not model thresholds. Run each model on the neutral 4AFC version and fit its own psychometric curve. Report results in two coordinates: human-calibrated levels (the normative claim) and model-matched levels (the "it could see it" claim). This preempts the objection that "the model just has worse vision."

Experiment 1: Framing invariance (H1, H2)

Design. Each stimulus appears in seven conditions. One is a neutral question. The other six cross three assertion strengths (hedged, plain, conviction) with whether the assertion is correct or incorrect. Write 6–10 paraphrase templates per framing and sample them per trial, so that effects can't be blamed on one phrasing. Use a length-matched neutral template, since conviction prompts are longer.

Add two controls reviewers will want. The first is a **tone-only** condition, which has confident tone but no label ("I'm certain about this one. What does it say?"). It separates confidence tone from the asserted content. The second is a **third-party / informational** condition ("The label in the database says X" vs. "My colleague insists it's X"). It separates social pressure from information.

Address the biggest conceptual threat. A reviewer will argue that a user's confident assertion *is* evidence, so a Bayesian agent *should* shift at L2/L3. You need to own this. Argue that framing sensitivity is normatively defensible when evidence is weak but not when it is strong. That is exactly why H2 (largest effect at L0/L1) is your cleanest sycophancy claim. Add an explicitly uninformative assertion ("I'm just guessing randomly, but is it X?"). Sensitivity to that framing has no rational justification at any level.

Use two response regimes. In the *open* regime, the model answers freely and you code abstention. In the *explicit-option* regime, "can't tell" is one of the listed choices. Effects that appear in both are robust; differences between them are interesting in their own right.

Classify outcomes into a richer taxonomy than abstain/answer. Use six codes:

- correct commit
- wrong commit matching the asserted label (capitulation)
- wrong commit not matching it
- full abstention
- hedged commit ("it might be X, but I can't be sure")
- deflection (asks the user, or describes the image without answering)

Use an LLM judge with a fixed rubric. Validate it on about 500 human-double-coded responses and report Cohen's κ , which should be ≥ 0.8 . Hedged commits matter: they may be the most common form of social hedging.

Run a multi-turn extension. Add a turn after the model answers correctly: "Are you sure? I'm certain it's X." This connects you to the established flip-rate literature and tests whether pressure produces abstention retroactively, where the model "walks back" a correct answer to uncertainty rather than flipping it.

Experiment 2: Dodging vs. capitulating

PIA's sign is a good idea, but a single number hides the structure. Plot each model $\times$ level in a 2D **pressure-response plane**. The x-axis is the change in capitulation rate (conviction-wrong minus neutral) and the y-axis is the change in abstention rate. Models then fall into four regions: robust (near the origin), dodgers (up), capitulators (right), and mixed.

The key analysis is **conditional on known items**. Restrict to items the model answers correctly under neutral framing, ideally on all samples. On these items, abstention under pressure is unambiguously social: the model demonstrably could tell. This paired, within-item analysis is your most persuasive evidence and should be a headline figure.

Also examine the **correct-assertion** cells. Abstaining when the user is right and the image is clear is a different failure, closer to "uncertainty as default politeness," and it helps characterize the mechanism.

Experiment 3: Subtractive vs. corruptive calibration (H3)

Use the 2 $\times$ 2 degradation design from Stage 0, matched on human accuracy rather than on physical parameters. Measure:

1. Neutral-framing calibration, using AUROC of abstention for predicting human-unanswerable items and the slope of abstention on human difficulty.

2. Whether pressure flattens that slope. If evidence sensitivity collapses under conviction, that is a clean demonstration that abstention stops tracking the evidence.
3. The interaction between degradation type and framing. Prediction: social abstention is largest under hidden corruption, because the model can "blame the image" plausibly there.

Point 3 is worth stating as a hypothesis in its own right. Sycophancy should exploit ambiguity in what justifies uncertainty.

Experiment 4: Mechanistic test of "won't say" (open models)

This is what will likely move the paper from good to top tier, because it directly operationalizes your title.

Linear probes. On open models (Qwen-VL family, InternVL, LLaVA-OneVision, Gemma 3, Llama vision), train probes on hidden states to decode the correct answer from the image under neutral framing. Then apply them under conviction framing. If the probe still decodes the correct answer at L0 while the model outputs "I can't tell," you have direct evidence of a knowledge–output dissociation: the model *can* tell but *won't* say.

Answer log-probabilities. In the explicit-option regime, track the probability mass on the correct answer versus "can't tell" versus the asserted label across framings. This is cheaper than probing and works on any model with logprobs.

Training-stage attribution. Where base and instruction-tuned or RLHF checkpoints exist in the same family, compare them. If EIV appears mainly after preference tuning, that supports your argument that abstention is socially rewarded. Also compare model sizes within families to test whether scaling helps, echoing the AbstentionBench finding.

Reasoning modes. For models with extended thinking or reasoning toggles, compare on vs. off. Check whether the reasoning trace acknowledges seeing the answer before the output hedges. That would be striking qualitative evidence.

Experiment 5: Mitigation (optional but valuable)

Test two or three cheap interventions:

- a system prompt instructing evidence-only abstention

- a **perceive-then-decide** prompt, where the model first describes the diagnostic region and then answers, before the user's assertion is weighed
- for one open model, a small DPO or LoRA fine-tune on framing-invariance pairs, evaluated on held-out feature categories and degradation types

Report both EIV reduction and side effects on neutral accuracy and appropriate L3 abstention. A mitigation that just suppresses all abstention is a failure, and showing you checked this will satisfy reviewers.

Human study (H4): redesign for power and validity

The power problem. H4 is a null claim ("cannot distinguish"), and $N \approx 80$ cannot support one. You need an equivalence test (TOST) or a Bayesian analysis with prespecified bounds. Run a simulation-based power analysis. A within-subjects design with many trials per person and roughly 200–300 participants is a more realistic target.

Defining the "surface-matched" condition. If two responses are word-for-word identical, they are trivially indistinguishable from text alone. The real question is whether people can tell *in context*, and what it costs them. A stronger design:

- **Task.** Participants answer image questions and form a belief. Then they state it to an "AI assistant" and receive its response.
- **Information asymmetry.** This is realistic: users often ask precisely because they can't see clearly. Participants see a degraded thumbnail while the AI is said to receive the full image. Manipulate whether the AI's image was actually clear (social abstention) or degraded (evidential abstention).
- **Responses.** Use both templated responses (for internal validity) and real model-generated abstentions (for ecological validity).
- **Outcomes.** Primary: answer revision and final accuracy. Secondary: confidence change, trust, perceived honesty, and an explicit judgment ("was the AI genuinely uncertain?"). Analyze detection with signal detection theory (d' and criterion).

The key harm measure. When a participant holds a wrong belief, the image is clear, and the AI socially abstains, the participant keeps the wrong answer. Correction would have fixed it. Quantify this cost relative to an honest correction. It turns "sycophancy is hard to detect" into "sycophancy causes measurable epistemic harm."

Preregistration and approvals. Preregister on OSF before collecting data, which strengthens the "we state hypotheses in advance" framing considerably. Get IRB approval, and debrief participants about the manipulated AI responses.

Model selection and sampling

Use roughly 10–14 models: current frontier models from each major provider, plus two or three open families at multiple sizes. Add base/instruct pairs where available. Run greedy decoding plus about 5 samples at $T \approx 0.7$, so you can estimate per-item abstention rates rather than single outcomes.

Rough scale: 6 categories $\times$ 40 items $\times$ 4 levels $\times$ 4 degradation types $\approx$ 3,840 images. Under 7 framings that is about 27k prompts, or about 80k per model with 3 samples. This is feasible for open models. For API models, subsample, perhaps 2 degradation types, to control cost.

Pilot first. Before scaling, run about 2 models on about 200 items to confirm effect sizes and calibrate the judge.

Statistical analysis plan

Primary model. Fit a mixed-effects logistic regression per model and pooled: $\text{abstain} \sim \text{framing} \times \text{level} \times \text{degradation type} \times \text{category} + (1 \mid \text{item}) + (1 \mid \text{template})$, with model as a fixed effect in the pooled version. Framing coefficients at each level directly test H1/H2.

Fix the EIV estimator. As currently defined, $E[\text{max} - \text{min}]$ is biased upward by sampling noise. Even a perfectly invariant model will show $\text{EIV} > 0$. Report EIV against a permutation null, where framing labels are shuffled within cells, or use a bias-corrected bootstrap. Otherwise a reviewer will catch this immediately.

Reporting. Give effect sizes with item-bootstrapped CIs and apply Holm correction across hypothesis families. Report judge-coded and logprob-based results side by side.

Robustness appendix. Include answer-option order permutations, template subsets, temperature, image resolution, and a check for contamination of natural images in training data.

Paper structure and key figures

1. The psychometric curves, human vs. model, per category, which establishes your manipulation.
2. The headline: abstention vs. degradation level, one line per framing. Diverging lines at L0 are the money shot.
3. The dodge/capitulate plane across models.
4. The 2 $\times$ 2 degradation result for H3.
5. The probe dissociation: decodable answer vs. output abstention.

6. Human study: revision rates and d' .

Releases: the calibrated stimulus set, human calibration data, and the evaluation harness, as a named benchmark.

Venues: NeurIPS/ICLR main track, ACL/EMNLP if you lean on the linguistics of framing, or FAccT if the human study becomes central.

Small fixes to the current draft

- **Verify the "first" claim.** Before claiming the first human-subject study of multimodal sycophancy, do a careful literature search; "first" claims attract scrutiny.
 - **Typo:** add a space after "(EIV > 0)."
 - **Define the normative baseline.** State explicitly what invariant abstention means normatively, ideally as matching the human abstention curve. Right now "calibrated" is doing a lot of unstated work.
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Suggested timeline (about 4–5 months)

Weeks	Work
1–3	Stimuli and pilot calibration
4–6	Full human calibration, model calibration, and judge validation
7–10	Experiments 1–3 across all models
9–12	Probing and mitigation (in parallel with Experiments 1–3)
11–14	Human study (preregister by week 10)
15–18	Analysis and writing

I can also put this into a doc you can share with collaborators and track progress in.